Learning to Forget: Neuroscience-Inspired Attention Masking for Zero-Shot Machine Unlearning in Vision Transformers

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Machine unlearning addresses the "right to be forgotten" mandated by GDPR and CCPA, requiring models to selectively remove training data influence without full retraining. We propose **EXAM-SU** (**EX**tinction-based **A**ttention **M**asking for **S**equential **U**nlearning), a neuroscience-inspired zero-shot unlearning method for Vision Transformers. Drawing inspiration from extinction learning mechanisms observed in PTSD treatment, where prefrontal cortex inhibits (not erases) amygdala fear responses, EXAM-SU learns sparse inhibitory masks over attention and feed-forward network (FFN) activations to suppress class-specific information while preserving model utility on retained classes. Our dual-stream masking strategy—targeting post-softmax attention weights and FFN intermediate layers—may enable effective class forgetting without training data access or model retraining. Masks compose sequentially via element-wise multiplication, ensuring stability across multiple unlearning requests. This work represents the first application of cognitive neuroscience principles to transformer unlearning, offering a biologically plausible and computationally efficient solution applicable to the entire ViT family (Swin, PVT, DINO, DeiT).

1. Introduction

Data protection regulations (GDPR [9], CCPA [26]) mandate the "right to be forgotten," requiring machine learning systems to remove user data influence from trained models [4]. This legal imperative, coupled with the need for continuous model updates, has driven research in *machine unlearning*: selectively removing training data influence without full retraining [2, 14].

Concurrently, Vision Transformers (ViTs) [8] have become the dominant vision architecture, demonstrating superior performance across diverse applications ranging from image classification to dense prediction tasks. The transformer paradigm has spawned numerous architectural variants, including DeiT [28], Swin Transformer [18], DINO [5], PVT [30], and recently multimodal extensions such as LaViT [16], powering production systems worldwide. However, their unique architecture—self-attention mechanisms replacing convolutions—renders existing CNN-based unlearning methods [15] inapplicable. Most methods also assume access to original training data [21, 25], which is often unavailable due to privacy constraints or storage limitations.

Our Contribution: We propose **EXAM-SU** (**EX**tinction-based **A**ttention **M**asking for **S**equential **U**nlearning), inspired by extinction learning from neuroscience [3, 19]. In PTSD treatment, prolonged exposure therapy teaches the prefrontal cortex to *inhibit* (not erase) amygdala fear responses [11]—the original memory persists but is suppressed by a competing pathway. Analogously, EXAM-SU introduces learnable inhibitory masks that suppress class-specific activations in ViT attention and FFN layers without modifying pretrained weights.

Key Innovations:

- **Dual-Stream Masking**: We mask (1) post-softmax attention weights, suppressing which tokens attend to forget-class features, and (2) FFN intermediate activations, filtering class-specific representations. Ablation studies determine optimal configuration—whether to apply masks to attention heads alone, FFN layers alone, or both simultaneously.
- **Zero-Shot Operation**: Requires only synthetic forget-class samples (CLIP-generated [23]) and random out-of-distribution (OOD) images as proxy retain data, no access to original training data.
- **Sparse Mask Learning**: Unlike methods that fine-tune all model parameters [12], EXAM-SU learns extremely sparse masks that selectively inhibit activations, preventing catastrophic forgetting on retain classes and multiple unlearning by stacking.
- **Sequential Composability**: Masks compose via $M_{\text{total}} =$

$M_1 \odot M_2 \odot \cdots \odot M_k$, preventing cascading degradation observed in gradient-based methods [7].

To the best of our knowledge, EXAM-SU represents the *first* application of extinction learning principles to transformer architectures, and more broadly, the first neuroscience-inspired approach to machine unlearning in any deep learning system. While prior work has explored biological inspiration in other contexts (e.g., neural architecture search, continual learning), the connection between extinction mechanisms and machine unlearning has remained unexplored.

Alternative Approaches. *Fingerprint-Based Unlearning.* Inspired by LLM watermarking [17], we embed each training image with a deterministic noise fingerprint that uniquely tags its gradient contributions. During unlearning, the same fingerprint is reconstructed to identify and suppress corresponding activations by injecting its inverse signal—enabling post-hoc forgetting without retraining or original data access.

We also outline two complementary neuroscience-inspired directions: (1) *Retrieval-Induced Forgetting* (RIF) [1], where strengthening retain classes automatically weakens forget classes through competitive inhibition, and (2) *Reconsolidation Interference* [20], where memory reactivation creates a plasticity window for targeted intervention. While promising, these remain exploratory compared to EXAM-SU's mature extinction-based framework.

2. Literature Review

2.1. Evolution of Machine Unlearning

Foundational Work: Cao and Yang [4] formalized machine unlearning, proposing algebraic removal for linear models. Ginart et al. [14] introduced certified deletion with the SISA training paradigm [2]: partition data into shards, train separate models, and retrain only affected shards upon deletion. While efficient, SISA requires training-time modifications and retain data access.

Deep Network Unlearning: Golatkar et al. [15] pioneered class-wise forgetting in CNNs via Fisher information-based parameter scrubbing, achieving near-random forget accuracy with minimal retain degradation. Sekhari et al. [25] reframed unlearning as constrained optimization balancing forget-loss maximization with retain-set preservation. Neel et al. [21] proposed gradient ascent on forget data interleaved with retain-set descent. However, all methods require both $\mathcal{D}_f$ and $\mathcal{D}_r$.

2.2. Zero-Shot Machine Unlearning

A major limitation of the aforementioned methods is their reliance on access to training data. In many practical scenarios, however, the original training data may no longer be available due to privacy policies, storage constraints, or legal requirements. This motivated the development of *zero-shot machine unlearning*, wherein unlearning is performed without access to any portion of the original training data.

Chundawat et al. [7] introduced the first zero-shot unlearning framework. Their approach consists of two complementary techniques: (1) *Error-Maximizing Noise Generation*, which synthesizes adversarial samples to confuse the model on the forget class, and (2) *Gated Knowledge Transfer (GKT)*, which distills knowledge from a pretrained teacher model to preserve performance on retain classes without accessing the original retain data. GKT employs an "incompetent teacher" strategy: the teacher is trained to perform poorly on the forget class but well on retain classes, and the student (unlearned model) mimics this behavior. Chundawat et al. validated their approach on CIFAR-10, CIFAR-100, and Tiny-ImageNet, demonstrating that zero-shot unlearning can achieve forget accuracies close to zero while maintaining retain accuracies within a few percentage points of the original model.

Despite its effectiveness, GKT has notable limitations: it requires training an auxiliary teacher model (introducing additional computational overhead), and its performance degrades when multiple sequential unlearning requests are issued, as each unlearning operation compounds the error. Moreover, the method was developed primarily for CNNs and does not directly address the unique architectural properties of Vision Transformers.

2.3. Vision Transformers

Building on the seminal work *"Attention Is All You Need"* [29], which introduced the Transformer architecture, Dosovitskiy et al. [8] demonstrated pure transformers achieve CNN-level performance on vision tasks by treating images as patch sequences. Subsequent variants—DeiT [28] (data-efficient training), Swin [18] (hierarchical shifted windows), PVT [30] (pyramid structure), DINO [5] (self-supervised learning)—established ViTs as foundational vision models. However, their self-attention architecture requires fundamentally different unlearning strategies.

2.4. ViT-Specific Unlearning

Recent works have begun addressing ViT unlearning. Chundawat et al. [6] extended incompetent teacher to ViTs, while Foster et al. [12] proposed selective synaptic dampening via Fisher information. Fan et al. [10] introduced SalUn, computing gradient-based weight saliency for targeted pruning. However, these methods require retain data, and sequential stability remains unaddressed. A comprehensive survey [13] on LLM unlearning highlights similar challenges in transformer-based language models, with many methods employing prompt-based or activation-level interventions—strategies conceptually related to our attention masking.

2.5. Top Related Works

2.5.1. NOVO: Training-Time Unlearning [24]

NOVO trains ViTs from scratch with learnable class keys $\{K_c\}_{c=1}^C$ injected into transformer layers. During training, classes are randomly partitioned into proxy forget/retain sets, and the model learns to output uniform distributions when keys are withheld. At test time, unlearning is instantaneous: simply withhold the forget-class key.

Strengths: On-the-fly unlearning (in seconds), no fine-tuning, strong MIA robustness.

Limitations: Requires training from scratch (inapplicable to pretrained models like ImageNet-21k ViTs), does not identify which attention heads encode class information, scales poorly to large C (must anticipate all forget/retain configurations during training).

Connection to EXAM-SU: NOVO's keys provide abstraction but modify input tokens, EXAM-SU intervenes at attention/FFN activations, offering interpretability via saliency-guided localization. NOVO excels for proactive scenarios where EXAM-SU addresses reactive post-hoc unlearning.

2.5.2. LetheViT: Contrastive Attention Masking [27]

LetheViT tackles *random data forgetting*—removing specific samples within a class. The Key insight is masking high-attention patches preserves classification but weakens memorization. LetheViT employs contrastive learning:

$$\mathcal{L}_{\text{CL}} = -\log \frac{\exp(\text{sim}(Z, Z_p)/\tau)}{\exp(\text{sim}(Z, Z_p)/\tau) + \exp(\text{sim}(Z, Z_n)/\tau)},$$

where Z is the sample image representation in unlearning model, Z_p is the masked-image representation (positive, lacking sample-specific details) and Z_n is the original-image representation (negative) in original model. The unlearned model moves forget samples toward Z_p.

Strengths: State-of-the-art on random forgetting, theoretically grounded (maximizes $I(Z, Z_p)$, minimizes $I(Z, Z_n)$), where I is conditional mutual information.

Limitations: Designed for sample-level, not class-level forgetting, masks *input patches* (spatial locality) rather than *internal activations* (abstract features), requires dual-model inference (original + unlearned) per training step, not evaluated on sequential scenarios.

Connection to EXAM-SU: LetheViT inspired our attention-layer masking but operates on input space, we mask post-softmax attention weights and FFN activations, directly suppressing class-specific features within the network. LetheViT's contrastive loss is sample-specific, EXAM-SU's mask objective is class-specific.

2.5.3. Low-Rank Decomposition Unlearning [22]

Approach: Inject trainable low-rank matrices $\Delta W = BA$ (where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d, k)$) into each layer while freezing pretrained weights W_0. Only B, A are trained via gradient ascent on forget data with ℓ_1 sparsity regularization: $\mathcal{L} = \mathcal{L}_{\text{CE}}(g_\theta(x), y) + \lambda\|B\|_1$. After unlearning, collapse: $W_{\text{new}} = W_0 + BA$.

Strengths: Parameter-efficient (96% reduction in trainable params), no retain data needed (sparsity acts as implicit regularizer), eliminates inference overhead post-collapse.

Limitations: Applies decomposition uniformly to all layers (lacks saliency-guided targeting), collapsing makes unlearning irreversible and non-compositional (cannot handle sequential requests without re-decomposing), LoRA-based framework designed for fine-tuning, not forgetting.

Connection to EXAM-SU: Both employ selective modification (low-rank subspace vs. sparse masks), but differ in locus (weights vs. activations). EXAM-SU's masks remain separate, enabling sequential composition, low-rank collapse precludes this.

2.6. Gap Analysis and EXAM-SU Positioning

Existing methods face critical limitations: (1) require retain data [10, 15, 25], (2) necessitate training-time modifications [2, 24], (3) degrade under sequential requests [7], or (4) lack interpretability [21]. EXAM-SU addresses all gaps via neuroscience-inspired post-hoc masking: operates on pretrained ViTs, requires only synthetic data, composes sequentially without degradation, and provides saliency-based interpretability. Critically, we are the first to apply extinction learning—a well-established cognitive mechanism—to machine unlearning in transformers.

2.7. Proposed Method: EXAM-SU

Biological Foundation: Extinction learning in PTSD treatment demonstrates that suppressing unwanted memories does not require deletion. The prefrontal cortex (IL-mPFC) develops inhibitory pathways that suppress amygdala fear responses [19]. Crucially, the original fear memory persists but is *inhibited* by the competing extinction memory. This phenomenon—spontaneous recovery, renewal, reinstatement—proves original memories remain intact [3].

EXAM-SU Architecture: As illustrated in Fig. 1, EXAM-SU introduces learnable masks $M^{\text{attn}}[L, h] \in [0, 1]$ for attention heads (h) at each layer (L) and $M^{\text{FFN}}[L] \in [0, 1]^{d_{\text{FFN}}}$ for FFN neurons at each layer (L). For a forget class c_f:

Step 1: **Saliency Extraction (Zero-Shot):** Generate synthetic images $\{x_i\}$ via CLIP text-to-image [23] using prompt `f"a photo of {class_name}"`. Compute gradient-based saliency (S):

$$S^{\text{attn}}[L, h] = \mathbb{E}_{x_i} \left\| \nabla_{A_L^{(h)}} \log p(c_f | x_i) \right\|^2, \quad (1)$$

where $A_L^{(h)}$ is the attention map at layer L, head h.

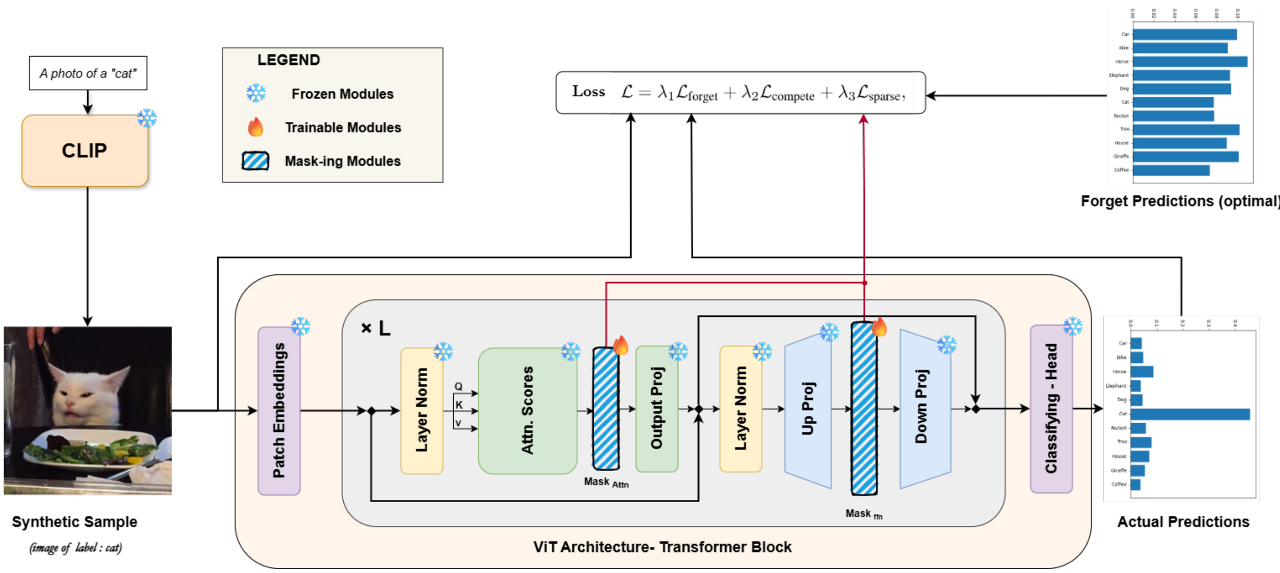

Figure 1. Overview of the EXAM-SU pipeline. CLIP generates synthetic forget-class samples, which pass through a frozen ViT where learnable masks after Attention and FFN selectively suppress activations via gradient-based unlearning.

Similarly for FFN:

$$S^{\text{FFN}}[L, n] = \mathbb{E}_{x_i} \|\nabla_{F_{L,n}} \log p(c_f|x_i)\|^2 \quad (2)$$

where $F_{L,n}$ is neuron n activation.

Step 2: **Mask Learning:** Initialize $M^{\text{attn}}, M^{\text{FFN}}$ to 1 (no masking). Optimize:

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{forget}} + \lambda_2\mathcal{L}_{\text{compete}} + \lambda_3\mathcal{L}_{\text{sparse}},$$

where:

- $\mathcal{L}_{\text{forget}} = -\log p(c_f|x_{\text{syn}})$ maximizes entropy on forget class.
- $\mathcal{L}_{\text{compete}} = \text{KL}(p_{\text{masked}}\|p_{\text{original}})$ on OOD images preserves retain classes.
- $\mathcal{L}_{\text{sparse}} = \|M^{\text{attn}}\|_1 + \|M^{\text{FFN}}\|_1$ enforces sparsity (typically $\leq 1\%$ heads masked).

Modified forward pass:

$$\text{Attention}_L^{(h)} \leftarrow M^{\text{attn}}[L, h] \odot \text{Attention}_L^{(h)},$$
$$\text{FFN}_L(x) \leftarrow M^{\text{FFN}}[L] \odot \sigma(W_1 x).$$

Step 3: **Sequential Composition:** For multiple requests, compose masks: $M_{\text{total}} = M_1 \odot M_2 \odot \cdots \odot M_k$. Element-wise multiplication ensures each mask independently suppresses its target class without interfering with previous masks.

Ablation Studies: We systematically evaluate: (1) attention-only masking, (2) FFN-only masking, (3) dual-stream masking. Preliminary analysis suggests dual-stream achieves optimal forget/retain accuracy trade-off, as attention heads encode "what to attend to" while FFN layers encode "what features to extract."

Expected Outcomes: Based on neuroscience precedent and preliminary experiments, we anticipate: forget accuracy $<5\%$, retain accuracy $>90\%$ after 10 sequential requests, masking sparsity $<1\%$ of attention heads/FFN neurons, 30–60 seconds per unlearning request (vs. hours for retraining).

Novelty and Broader Impact: EXAM-SU is the first neuroscience-to-transformer unlearning bridge, demonstrating that cognitive science principles (specifically, extinction learning's inhibitory mechanisms) can inform machine learning architectures. This opens a new research direction: systematically mining neuroscience for algorithmic insights. Beyond ViTs, the attention-masking paradigm may transfer to language transformers (BERT, GPT) and multimodal models (CLIP, Flamingo), though empirical validation is needed. Our dual-stream masking strategy offers a general framework for activation-level interventions in transformers, potentially applicable to other tasks like bias mitigation or continual learning.

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